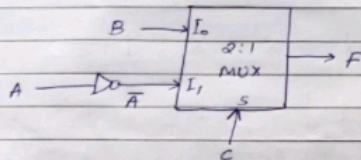


Assignment 3 - Combinational

1. Implement the following functions using
i) 2:1 MUX and NOT gates ii) 4:1 MUX
a) $F(A, B, C) = \overline{A}C + B\overline{C}$

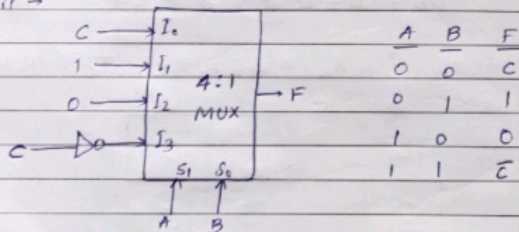
i.e.



When $C=0$, $F=B \Rightarrow I_0=B$

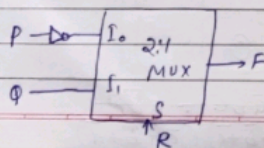
$C=1$, $F=\overline{A} \Rightarrow I_1=\overline{A}$

ii.e.



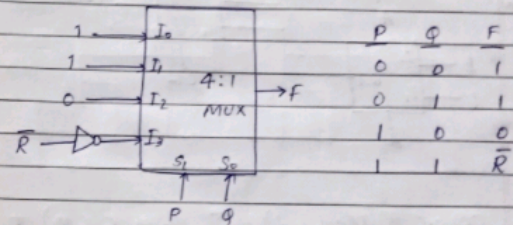
b) $F(P, Q, R) = \overline{P}\overline{R} + QR + P\overline{Q}$

i.e.



When $R=0$, $F=\overline{P}+Q$
 $R=1$, $F=P\overline{Q}$

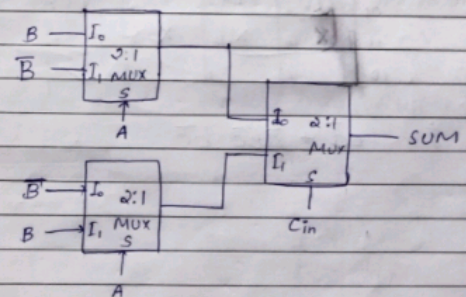
ii.e.

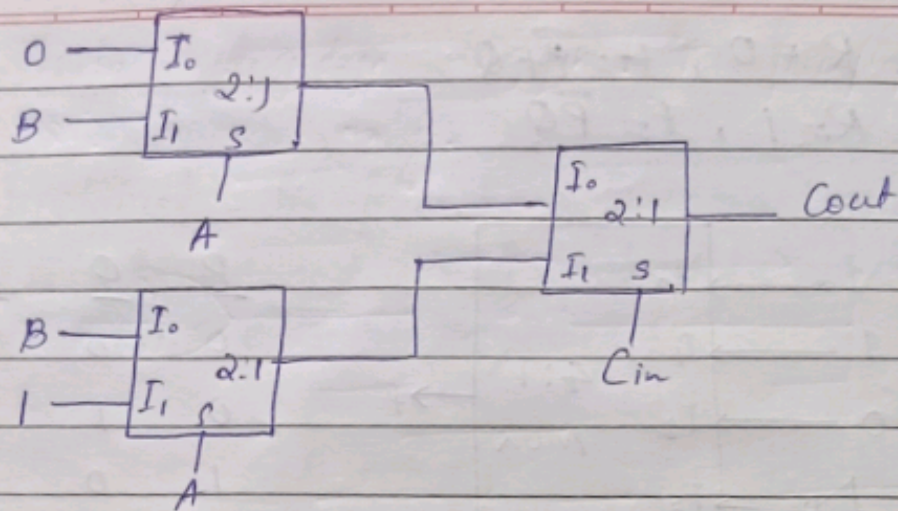


2. Implement the functionality of a full adder using 2:1 Muxes only.

Sum = $A \oplus B \oplus C_{in}$

Carry = $AB + BC + AC_{in}$





3. Simplify $f(a,b,c,d) = \sum m(2,4,5,6,10) + D(12,13,14,15)$ and implement the SOP expression using 4:1 Mux.

ab \ cd					
		00	01	11	10
00	0	0	1	3	2
01	4	1	5	7	6
11	12	X	X	X	14
10	8		9	11	10

$$f = cd + \cancel{abc} b\bar{c}$$